

CS5275 The Algorithm Designer's Toolkit

S2 AY2025/26

Tutorial problem set 4

(There is no need to submit your solution for tutorial problem sets)

Problem 1 (Concentration of Fourier spectrum). Suppose the Fourier spectrum of $f : \{-1, 1\}^n \rightarrow \mathbb{R}$ is ϵ -concentrated on $\mathcal{F}$. Let $g : \{-1, 1\}^n \rightarrow \mathbb{R}$ be a function satisfying $\|f - g\|_2^2 \leq \epsilon$. Prove that the Fourier spectrum of g is $O(\epsilon)$ -concentrated on $\mathcal{F}$.

Problem 2 (Pseudorandom generator via one-way permutation). Let $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ be a permutation. Suppose there exists a efficient deterministic algorithm A and a parameter $\gamma > 0$ such that

$$\Pr_{\mathbf{r}, \mathbf{s} \sim \mathbb{F}_2^n} [A(\mathbf{r}, f(\mathbf{s})) = \mathbf{r} \cdot \mathbf{s}] \geq \frac{1}{2} + \gamma,$$

where $\mathbf{r} \cdot \mathbf{s} = \sum_{i=1}^n \mathbf{r}_i \mathbf{s}_i \pmod{2}$ denotes the inner product over $\mathbb{F}_2$.

Show that one can use A together with the Goldreich–Levin algorithm to recover s from $f(s)$ for a $\gamma/2$ fraction of $s \in \mathbb{F}_2^n$ with non-negligible probability.

More precisely, prove the following steps.

1. Show that for at least a $\gamma/2$ fraction of strings $s \in \mathbb{F}_2^n$,

$$\Pr_{\mathbf{r} \sim \mathbb{F}_2^n} [A(\mathbf{r}, f(s)) = \mathbf{r} \cdot s] \geq \frac{1}{2} + \frac{\gamma}{2}.$$

2. Fix any such s in the subsequent discussion. Define

$$h_s(r) := (-1)^{A(r, f(s))}, \quad r \in \mathbb{F}_2^n.$$

Let

$$S := \{i \in [n] : s_i = 1\}.$$

Viewing h_s as a Boolean function on $\{-1, 1\}^n$, prove that

$$\widehat{h_s}(S) \geq \gamma.$$

3. Explain how to use the Goldreich–Levin algorithm to efficiently recover s from $f(s)$ with high probability.

Problem 3 (Multiplicative weights with exponential updates). Consider a variant of the multiplicative weights algorithm with exponential updates, with n experts and T rounds. On each round $t \in \{1, 2, \dots, T\}$:

1. The algorithm maintains nonnegative weights $w_1^{(t)}, \dots, w_n^{(t)}$ on the experts.

2. Let

$$\Phi^{(t)} := \sum_{i=1}^n w_i^{(t)}.$$

The algorithm chooses the distribution

$$p_i^{(t)} := \frac{w_i^{(t)}}{\Phi^{(t)}} \quad \text{for each } i \in [n].$$

3. Then a cost vector

$$\mathbf{m}^{(t)} = (m_1^{(t)}, \dots, m_n^{(t)}) \in [-1, 1]^n$$

is revealed.

4. The algorithm incurs expected cost

$$\mathbf{m}^{(t)} \cdot \mathbf{p}^{(t)} = \sum_{i=1}^n p_i^{(t)} m_i^{(t)}.$$

5. The weights are updated according to the exponential rule

$$w_i^{(t+1)} = w_i^{(t)} e^{-\eta m_i^{(t)}} \quad \text{for all } i \in [n],$$

where $0 < \eta \leq 1$ is a fixed parameter.

Assume the initial weights are uniform:

$$w_i^{(1)} = 1 \quad \text{for all } i \in [n].$$

Prove that for every expert $i \in [n]$,

$$\sum_{t=1}^T \mathbf{m}^{(t)} \cdot \mathbf{p}^{(t)} \leq \sum_{t=1}^T m_i^{(t)} + \eta \sum_{t=1}^T (\mathbf{m}^{(t)})^2 \cdot \mathbf{p}^{(t)} + \frac{\ln n}{\eta},$$

where

$$(\mathbf{m}^{(t)})^2 := ((m_1^{(t)})^2, \dots, (m_n^{(t)})^2).$$

Hint. Use the inequality

$$e^{-\eta x} \leq 1 - \eta x + \eta^2 x^2 \quad \text{when} \quad |\eta x| \leq 1.$$